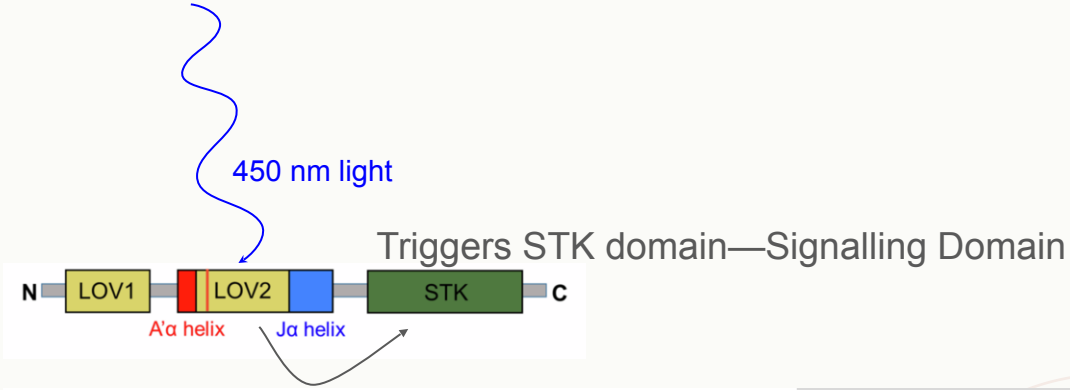
Structural Characterization of A Magnetically Sensitive Fluorescent Protein

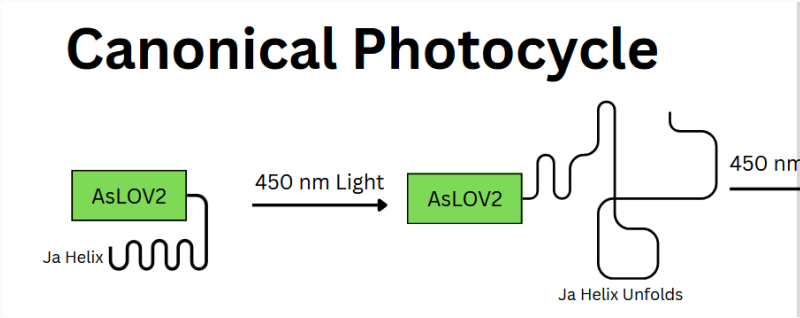
A presentation by Jerome Wu
Mentored by Dr Samsuzzoha Mondal and Professor Soichi Wakatsuki

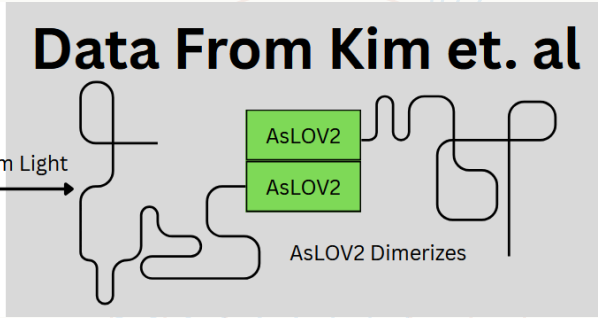


AsLOV2 is a Native Photoreceptor

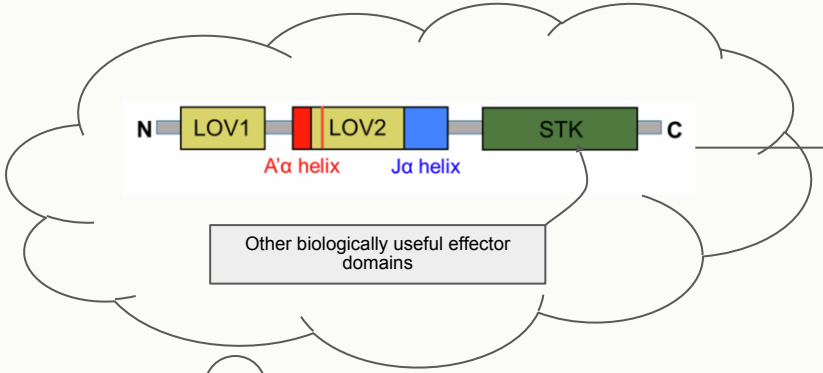




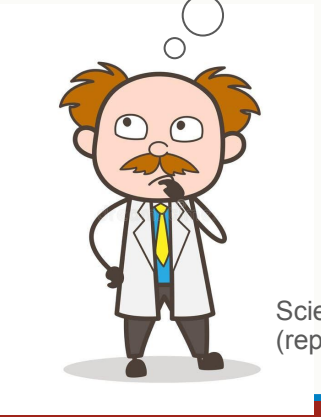




That Structural Change is used in Optogenetics



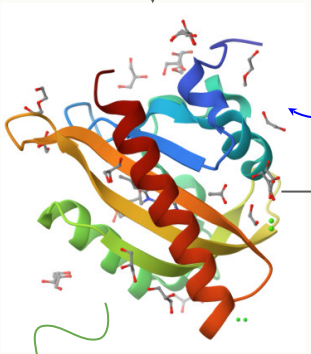
LOV-system	Effectors	Fold activation	Purpose	Lifetime (s)	References
AsLOV2	**Phototropism**			**2–4300 s (*WT* = 55 – 81)**	**a**
LINuSs1	NLS	3–7	Nuclear localization[20]	240 s (4 min)	Yazawa et al., 2009
LOV-TAP	TrpR	6, 70	DNA-binding/Tryptophan repressor	NA	Strickland et al., 2008; Wu et al., 2011
LOV-Rac	Rac1 (GTPase)	10	Control actin cytoskeletal dynamics	43 s	Wu et al., 2009
Tulips	ePDZ	2–49	Peptide caging		Strickland et al., 2012
LOV-TetR	TetR	NA	Tetracycline/DNA-binding	30 s	Moon et al., 2014



Scientists (representation)

MagLOV





450 nm light

Green light

mutations

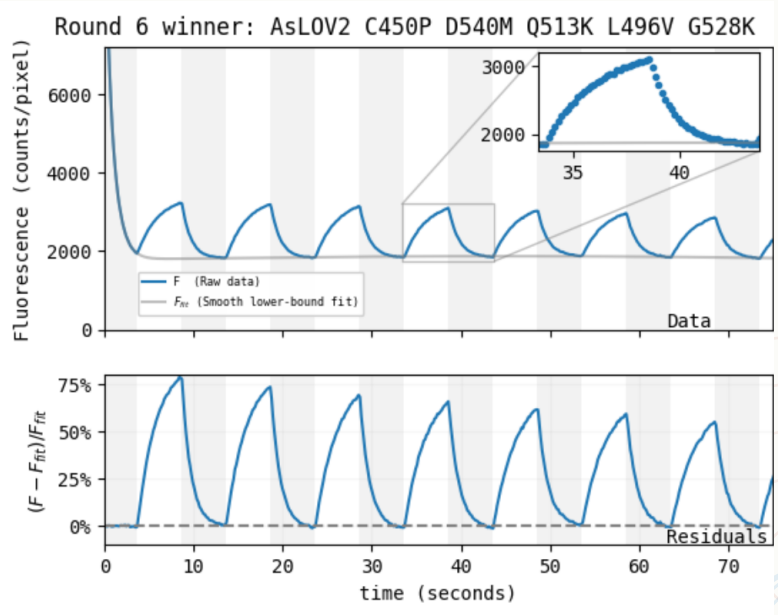
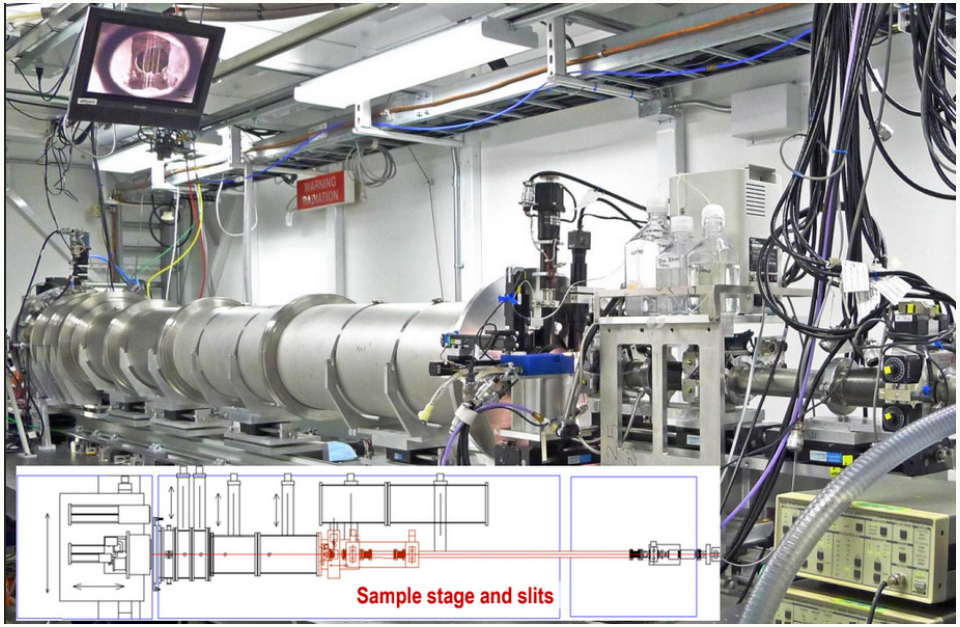


Figure from doi:10.5281/zenodo.8137174

Research Question: Does MagLOV undergo structural change in response to a magnetic field, and if it does then what is that structural change?

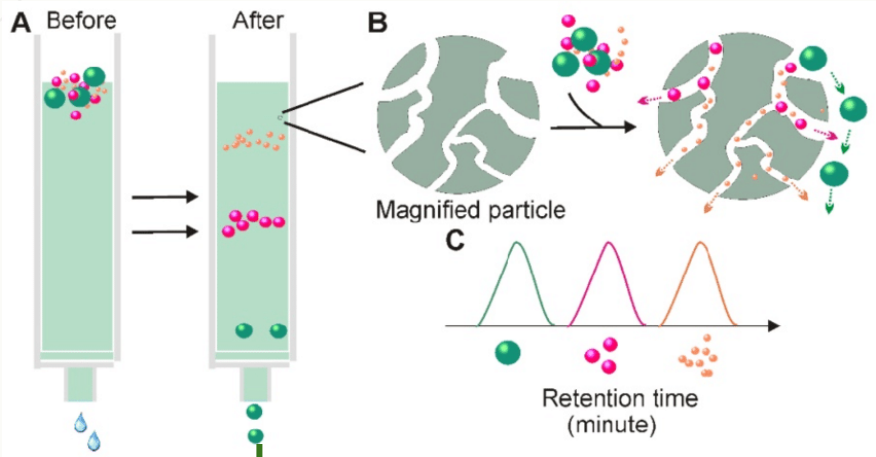
SEC-SAXS



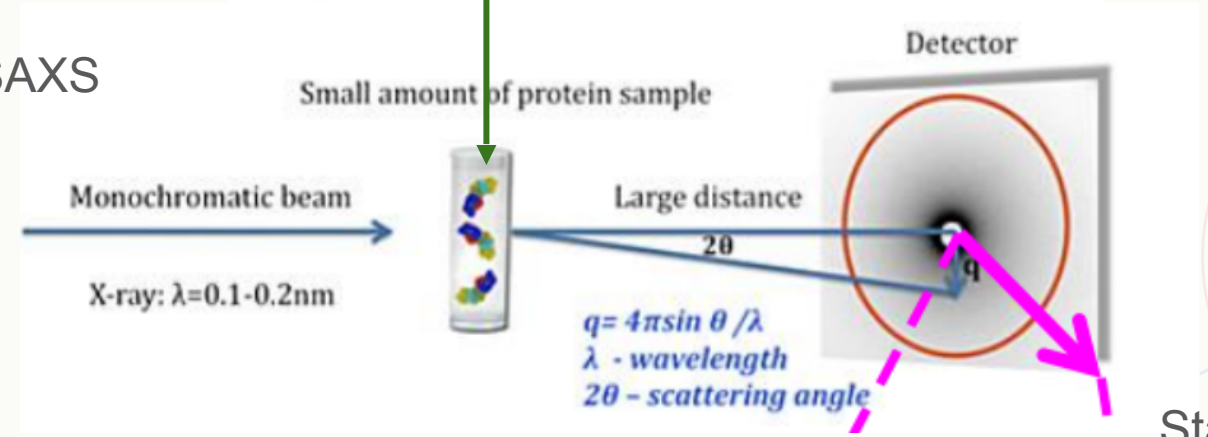
Thanks Tsutomu and The SAXS Team!

But what is SEC-SAXS?

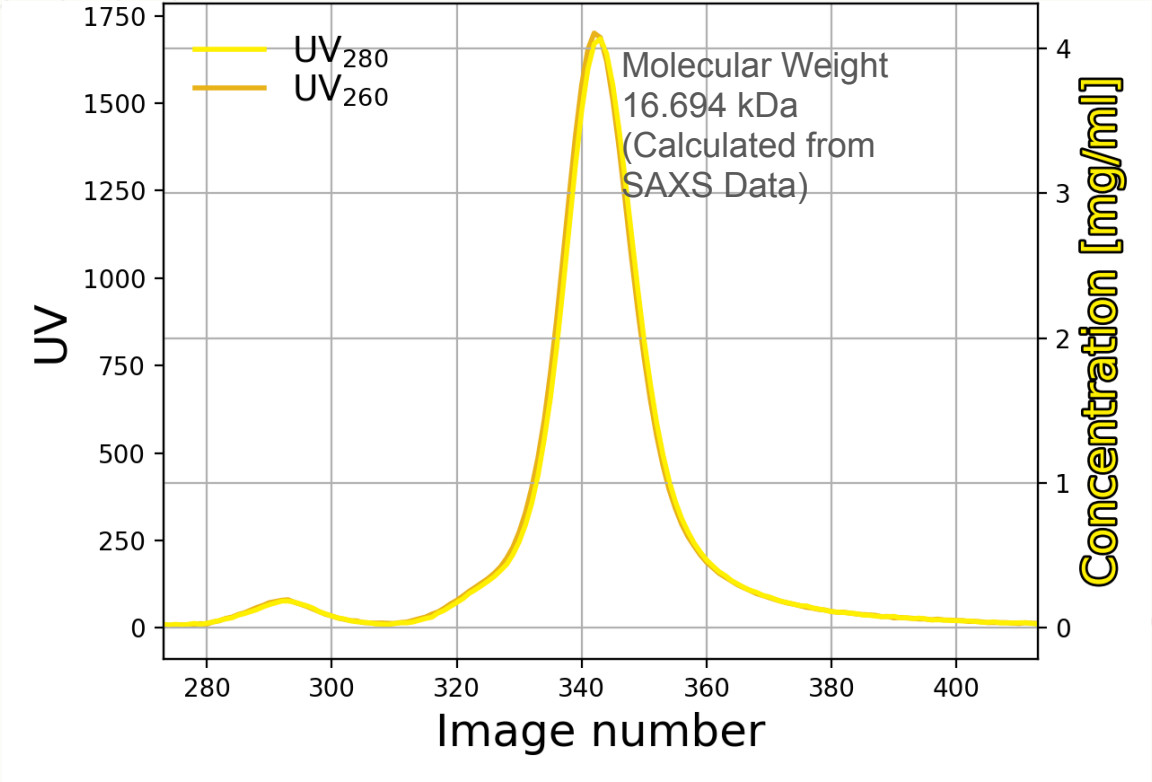
SEC



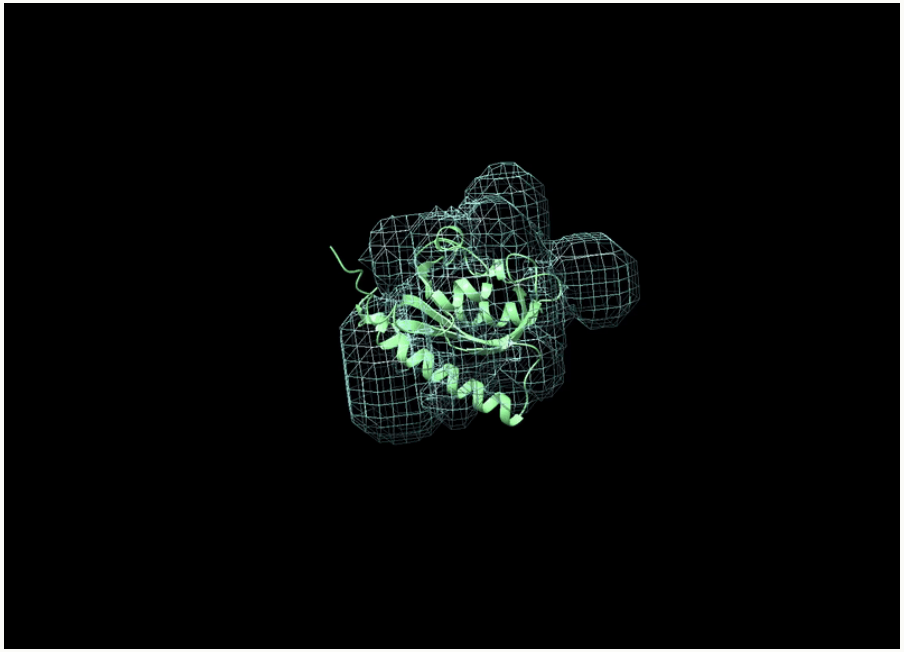
SAXS



SEC Trace


UV280
UV260
Molecular Weight
16.694 kDa
(Calculated from
SAXS Data)
UV
Concentration [mg/ml]
Image number
1750
1500
1250
1000
750
500
250
0
4
3
2
1
0
280
300
320
340
360
380
400

Ab Initio Models



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